SIH26189 – Skills Required

AI-Powered Criminal Network Analysis System

You do not need expert-level knowledge of every technology. Divide the project across the team. The main focus is Web Development + Graph Database + NLP + Graph Analytics.

Skill	Importance	Level
React / Frontend	★★★★	Good
Node.js / Backend APIs	★★★★	Good
Database	★★★★	Good
Neo4j / Graph DB	★★★★★	Basic–Intermediate
Python	★★★★	Basic–Intermediate
NLP / AI	★★★★	Basic
Graph Algorithms	★★★★	Basic
PDF/CSV Processing	★★★	Basic
Authentication	★★★	Basic
Git/GitHub	★★★	Good

1. Frontend – React

Know API integration, forms, tables, charts, search/filter, authentication UI and graph visualization. Cytoscape.js is a good choice for the investigation graph.

2. Backend

Node.js/Express is sufficient. Understand REST APIs and how the backend connects the frontend with Neo4j and other services.

3. Neo4j – Most Important New Skill

Learn nodes, relationships, properties and basic Cypher. Example: Person USES Phone, Person OWNS BankAccount, BankAccount TRANSFERRED_TO BankAccount. Multi-hop queries are central.

4. Python

One member should know Python basics, JSON/file handling, pandas and simple data processing for the NLP/data pipeline.

5. NLP / AI

Learn Named Entity Recognition (NER) to extract PERSON, PHONE, MONEY, BANK ACCOUNT and similar entities from FIR text. No need to train an AI model from scratch.

6. Entity Resolution

Understand name similarity, fuzzy matching, shared identifiers and confidence scores. Possible matches should preferably require investigator approval.

7. Graph Algorithms

Know Degree Centrality, Betweenness Centrality and PageRank at a basic level. These identify highly connected/central entities, not proof of criminality.

8. Transaction Analysis

Process CSV transactions and trace money flows such as Bank A → Bank B → Bank C → Crypto Wallet. Basic pandas and Neo4j knowledge is enough.

9. PDF Processing

Pipeline: PDF → text → NLP → entities → Neo4j. Useful Python libraries include PyMuPDF/pdfplumber and spaCy.

10. Security

Know authentication, role-based access, password hashing, HTTPS, audit logs, input/file validation and SHA-256. Suggested roles: Admin, Investigator, Viewer.

11. Data Structures

Be comfortable with JSON, lists/dictionaries and the graph model: Nodes + Edges + Properties.

12. Git/GitHub

Use branches, pull requests, merges and basic team collaboration.

13. Recommended Team Division

Member	Responsibility	Skills
1	Frontend	React, Tailwind, Cytoscape.js
2	Backend	Node.js, Express, APIs, Auth
3	AI/NLP	Python, NER, Entity Resolution
4	Graph	Neo4j, Cypher, Graph Algorithms
5	Data/Analytics	Pandas, Transactions, Centrality, Synthetic Data
6	Integration	Integration, testing, PPT, demo, documentation

14. What You Do NOT Need to Learn Deeply

• Deep Learning from scratch

• Neural-network mathematics

• Advanced TensorFlow/PyTorch

• Advanced statistics

• Blockchain development

• Advanced cybersecurity

• Training a huge LLM from scratch

• Computer vision

15. Personal Learning Order

1. Neo4j + Cypher
2. Graph concepts
3. Cytoscape.js
4. Node.js API
5. Python basics
6. NLP / NER
7. Entity Resolution
8. Transaction analysis
9. Authentication + audit
10. Full integration

Key Takeaway

If you already know React + Node.js, focus first on Neo4j, Cypher, Cytoscape.js, NLP/NER, entity resolution and graph algorithms.